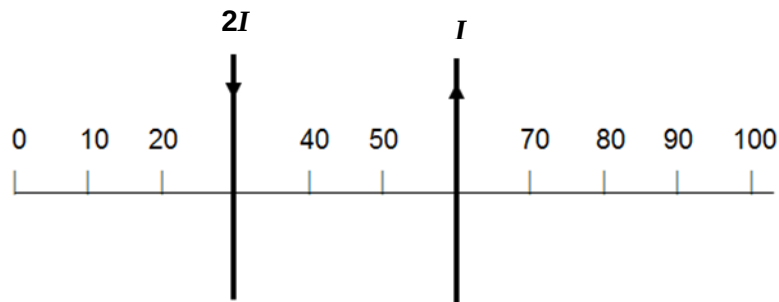


If students pick A, they have ignored the resistance of the voltmeter.

- 24 Two parallel, straight conductors, placed at right angles to a ruler, carry currents of $2I$ and I flowing in opposite directions.

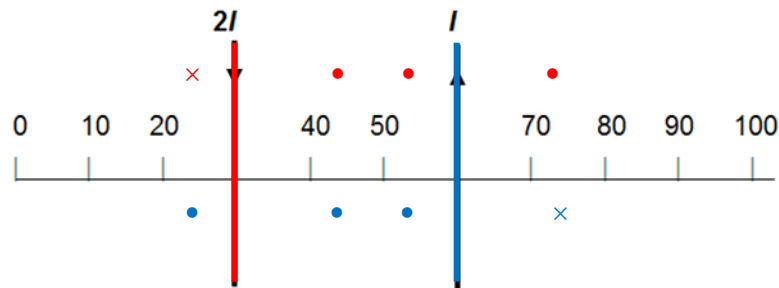


At which point on the ruler is the resultant magnetic field the greatest?

- | | |
|---|------------|
| A | 20 cm mark |
| B | 40 cm mark |
| C | 50 cm mark |
| D | 70 cm mark |

Answer: B

Using the RHGR, the magnetic field due to the individual wires are indicated on the diagram.



$$B_{40\text{cm}} = \frac{\mu_0(2I)}{2\pi(0.1)} + \frac{\mu_0(I)}{2\pi(0.2)} = 12.5 \frac{\mu_0 I}{\pi}$$

$$B_{50\text{cm}} = \frac{\mu_0(2I)}{2\pi(0.2)} + \frac{\mu_0(I)}{2\pi(0.1)} = 10 \frac{\mu_0 I}{\pi}$$

- 25 A beam of electrons, travelling horizontally towards the right, passes between two horizontal